Refund history: what the data shows

achopikyan · analysis of `data/history.csv` (251 rows, 248 unique requests, 1 June to 18 August 2026) and `data/pending.csv` (21 rows, 20 requests). Counts are recorded observations, not causal estimates.

Key findings

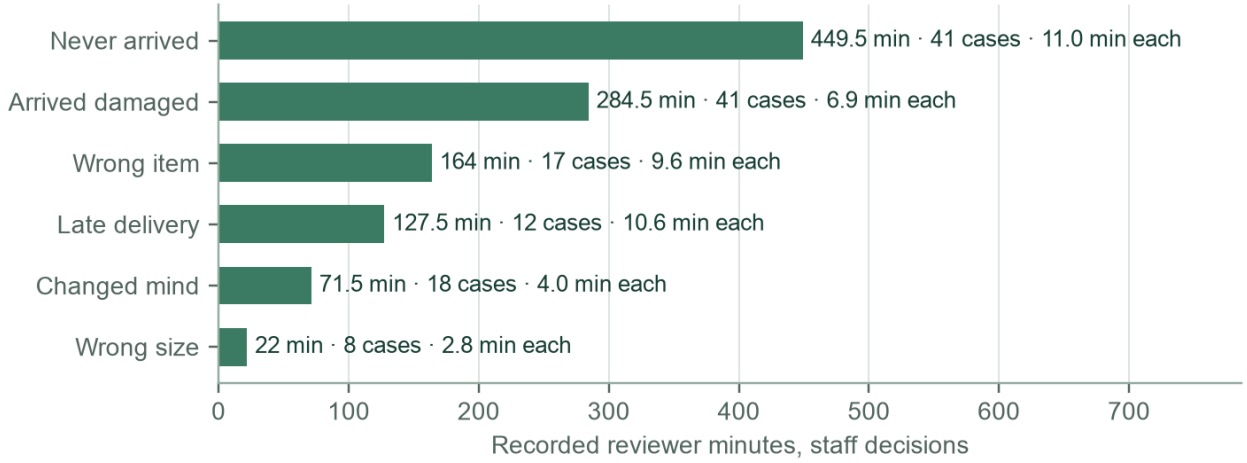
- Reviewers recorded **1,119 minutes** on 137 requests. Never-arrived claims took the most time; wrong-size returns the least.
- **8 approvals ended in a chargeback**: $2,319 refunded plus $200 in fees. 7 were reviewer approvals made against evidence already in the notes.
- Decisions are inconsistent: 4 change-of-mind refunds were approved after the 14-day limit, all by one reviewer, while 3 others were refused for the same reason. 3 rejections were overturned on appeal.
- The old rule is **amount below $55, tenure at least 9 months, no prior refunds, clear fraud flag** and the 14-day window. It explains 242 of 243 decidable requests, and it never reads notes.
- Only 34 of 248 requests have a recorded outcome, so no approach can claim a measured error rate yet.

1. The sample

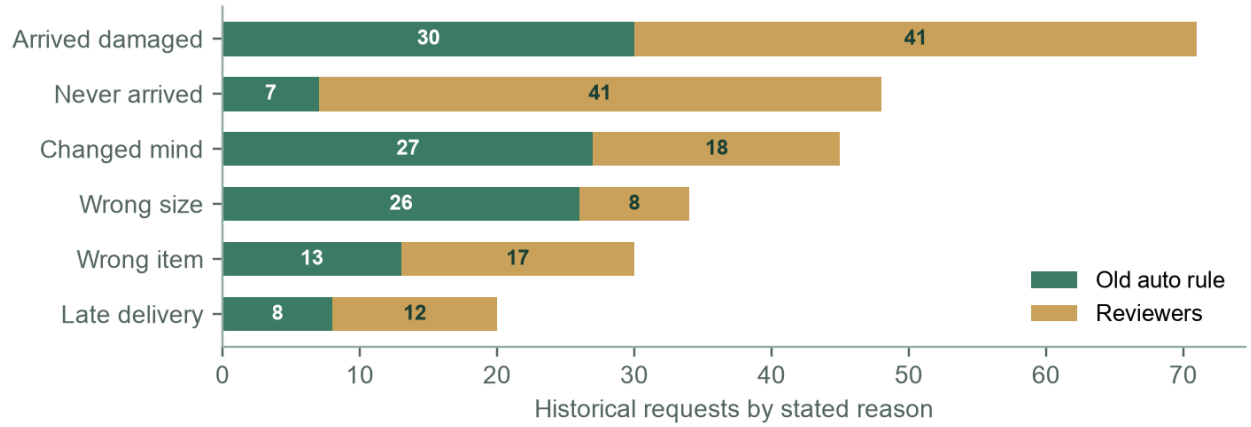
Measure	Value
Unique historical requests	248 (3 exact duplicate rows removed)
Channel	66 web form (RF-1xxx) · 182 support desk (RF-2xxx)
Decided by	old auto rule 111 · 4 reviewers 137
Decisions	215 approved · 9 partial · 24 rejected; the old rule only approves
Approved dollars	old rule $3,377.13 · reviewers $15,833.57, plus 9 partial refunds
Recorded outcomes	34 of 248; 214 blank

2. Where reviewer time goes

The brief says most staff reviews do not need to be read. The time data supports that: simple returns take a few minutes, while delivery disputes and late deliveries take ten or more, because they need evidence weighed against policy note 4.



The old rule already takes most wrong-size and change-of-mind requests; reviewers spend their time on disputes and damage:



3. Where money goes: chargebacks

Policy note 7: every chargeback costs the order amount plus a $25 fee. The refund amount stands in for the missing order amount. 7 of 8 chargebacks followed a reviewer approval where the notes already pointed the other way:

Request	Refund	Reason	Approved by	What the notes already said
RF-1065	$495	Arrived damaged	Reviewer	Fraud flag, new account, first order, no photos.
RF-1005	$410	Never arrived	Reviewer	Carrier proof of delivery, signed for at the address.
RF-2111	$390	Arrived damaged	Reviewer	Fraud flag, new account, no photos after two requests.
RF-1027	$340	Arrived damaged	Reviewer	Photos show the original packaging still sealed.
RF-2042	$265	Never arrived	Reviewer	Tracking shows delivered and signed.
RF-1045	$190	Never arrived	Reviewer	Customer told support the parcel did arrive, late.
RF-2083	$180	Arrived damaged	Reviewer	Customer said the item broke during self-assembly.
RF-2074	$49	Arrived damaged	Old rule	No notes; inside the old rule's limits.
Total	$2,319			plus $200 in fees: $2,519

Recorded chargebacks among approved or partial refunds, excluding one row with invalid data. These are counts, not calibrated rates: most outcomes are unknown and the groups are small.

Refunded cohort	Chargebacks / refunds
Requested amount at most $50	1 / 130
Over $50 through $100	0 / 36
Over $100 through $200	2 / 23
Over $200 through $300	1 / 22
Over $300	4 / 12
Customer tenure at least 24 months	0 / 95

4. Where decisions break

- **Change of mind after 14 days.** Note 2 says *no exceptions*, yet 4 requests were approved 18–27 days after delivery (RF-2012, RF-2055, RF-1034, RF-2105), all by one reviewer. 3 others were refused as outside the window.
- **Appeals.** RF-1008, RF-2090, RF-2143 were rejected and then refunded in full on appeal; one customer closed their account and one turned out to be a confirmed picking error, which note 6 refunds with no questions asked.
- **Damage without a photo.** Note 3 requires a photo above $50, but the old rule approved 3 damage claims of $51.26–$53 (RF-2037, RF-2046, RF-2121) with no photo in the notes.
- **Fraud confirmation arrives later.** 19 requests carried a fraud flag. For 11, the outcome note says the fraud team confirmed the card was stolen, after the refund decision. A rule that pays only confirmed stolen-card cases therefore matches none of them at decision time, which the Backtest shows.

5. The old rule, reconstructed

Policy note 1: *small stuff from good customers goes through automatically; IT set that up last year.* Searching the historical routing gives amount below $55, tenure at least 9 months, no prior refunds, a clear fraud flag and the 14-day change-of-mind window. It matches 242 of 243 requests with complete data; the exception is RF-1002. 5 rows with missing tenure or delivery days, negative days or a blank fraud flag are left out.

Applied to the pending queue, the same rule pays 6 requests totalling $233 (RF-1069, RF-1074, RF-1075, RF-2196, RF-2198, RF-2200). It cannot see that RF-1074's note (in Spanish) says the store already refunded the customer, that RF-1075 is a change of mind on day 22 filed as damage, or that RF-2196, RF-2198 and RF-2200 are caught in a Northfield carrier disruption.

6. Messy data and how the application handles it

Issue	In the files	Handling
Duplicates	3 exact duplicate history rows; RF-2183 appears twice in pending at $49 and $52	Exact duplicates collapse; conflicting versions are kept and must be reconciled before approval
Amount formats	USD 120.00, USD 24.06, $43, $36, 55.00 USD	Parsed to integer cents
Dates	9 non-ISO dates, day-first and ISO timestamps	Day-first by default, with a visible warning; timestamps reduced to dates
Fraud flag	no 226 · yes 16 · Y 4 · false 3 · N 1 · blank 1	yes/Y/true and no/N/false normalized; blank stays unknown and cannot pass a fraud-flag condition
Reason and category	arrived damaged, changed mind, damaged, late, wrong item, Electronics, Home	Aliases normalized to the dictionary values
Missing or invalid numbers	Tenure missing twice; delivery days missing once and negative once	Unknown values never satisfy a rule; negative days are a data error that sends the request for information
Reason contradicts notes	For example RF-1075: damage selected, note says it is fine	The LLM flags the conflict with a quote; the request goes to Review

7. What this means for the line

- Automate the cases that were cheap and safe in history (simple returns, small damage with a photo, confirmed merchant or fraud-team outcomes), and keep prior refunds, delivery disputes, electronics damage and anything over $300 with a person.
- Read the notes. Every chargeback above and most broken decisions were visible in the free text, which is why evidence extraction uses an LLM with exact quotes while the rules and payments stay in code.
- Test every change against this history before it goes live. On the same 248 requests, the application's Backtest gives:

Version	Automatic	Proposed refunds	Above human amount	Known chargebacks on automatic
v1 (old rule)	108	$3,332.51	$0 on 1 assessed	1
v2 (proposed)	108	$4,430	$0 on 32 assessed	0
v3 (broader)	125	$6,681.45	$415 in 3 of 44 assessed	0

Limits: 214 of 248 outcomes are blank, historical decisions can be wrong, the pending queue was inspected during development and is not a blind holdout, and refund amounts stand in for order amounts. Zero observed excess is not proof of safety.